AN INVERSE LIMIT OF QUASI-COMPACT SPACES IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of the limit as an intersection of closed subsets of the product cut out by the graphs of the transition maps. Used inside the counted proof (source lines 2700-2708). Presents the limit as a closed subset of the product.
- The statement that a closed subset of a quasi-compact space is quasi-compact. Invoked at the end of the counted proof (source line 2714). Concludes quasi-compactness.

Source reference. Original lemma lemma-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1879–1879. Navigation pointer only: the counted text refers to the Stacks item lemma-closed-in-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-graph-closed: sources/stacks-topology/pinned-original-source.tex, source lines 365–365. Navigation pointer only: the counted text refers to the Stacks item lemma-graph-closed, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original theorem theorem-tychonov: sources/stacks-topology/pinned-original-source.tex, source lines 2663–2663. Navigation pointer only: the counted text refers to the Stacks item theorem-tychonov, which is an existing labeled result

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Lemma 0.1. *Let $\mathcal{I}$ be a category and let $i \mapsto X_i$ be a diagram over $\mathcal{I}$ in the category of topological spaces. If each X_i is quasi-compact and Hausdorff, then $\lim X_i$ is quasi-compact.*

Proof. Recall that $\lim X_i$ is a subspace of $\prod X_i$. By Theorem theorem-tychonov this product is quasi-compact. Hence it suffices to show that $\lim X_i$ is a closed subspace of $\prod X_i$ (Lemma lemma-closed-in-quasi-compact). If $\varphi : j \to k$ is a morphism of $\mathcal{I}$, then let $\Gamma_\varphi \subset X_j \times X_k$ denote the graph of the corresponding continuous map $X_j \to X_k$. By Lemma lemma-graph-closed this graph is closed. It is clear that $\lim X_i$ is the intersection of the closed subsets

$$\Gamma_\varphi \times \prod\nolimits_{l \neq j,k} X_l \subset \prod X_i$$

Thus the result follows. □